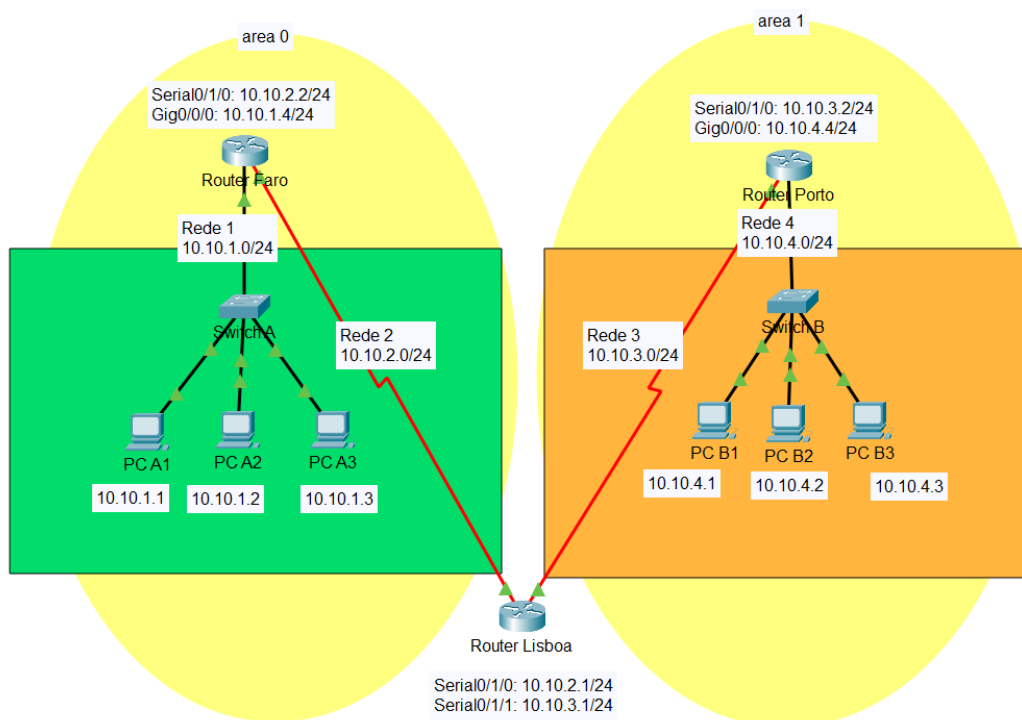


Context

In this activity, we will proceed with the basic configuration of OSPF in multiple areas, followed by the configuration of various types of areas, as well as the implementation of virtual circuits.

1 – Basic topology to be implemented in Packet Tracer

The packet tracer file provided in the attachment is already initialized with the addressing shown in the figure. This topology can be obtained from the link, where you should confirm the settings already made.



2 – OSPF Configuration

The OSPF configuration to be implemented contains two areas, but is similar to that of a single area. First, the protocol is activated, and then the OSPF areas are configured with the interfaces that are to be part of the OSPF process.

Faro router configuration

We start by configuring OSPF on the Faro router with connections to networks 10.10.1.0/24 and 10.10.2.0/24.

```
Router-Faro> en
Router-Faro# conf t
Router-Faro (config)# router ospf 100
Router-Faro (config-router)# network 10.10.1.0 0.0.0.255 area 0
Router-Faro (config-router)# network 10.10.2.0 0.0.0.255 area 0
Router-Faro (config-router)# ^Z
```

Porto router configuration

Next, configure the Porto router, which follows a similar structure.

```
Router-Porto> en
Router-Porto # conf t
Router-Porto (config)# router ospf 100
Router-Porto (config-router)# network 10.10.3.0 0.0.0.255 area 1
Router-Porto (config-router)# network 10.10.4.0 0.0.0.255 area 1
Router-Porto (config-router)# ^Z
```

Lisboa router configuration

And finally, the Lisbon router, where the only difference from the others is that it has an interface with each of the areas.

```
Router-Lisboa > en
Router-Lisboa # conf t
Router-Lisboa (config)# router ospf 100
Router-Lisboa (config-router)# network 10.10.2.0 0.0.0.255 area 0
Router-Lisboa (config-router)# network 10.10.3.0 0.0.0.255 area 1
Router- Lisboa (config-router)# ^Z
```

3 – Verification of routing tables

Using the show ip route command, check the resulting routing tables, denoting the routes learned by OSPF with the code "O."

Router-Faro# show ip route

```
10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C    10.10.1.0/24 is directly connected, GigabitEthernet0/0/0
L    10.10.1.4/32 is directly connected, GigabitEthernet0/0/0
C    10.10.2.0/24 is directly connected, Serial0/1/0
L    10.10.2.2/32 is directly connected, Serial0/1/0
O IA  10.10.3.0/24 [110/128] via 10.10.2.1, 00:01:15, Serial0/1/0
O IA  10.10.4.0/24 [110/129] via 10.10.2.1, 00:01:05, Serial0/1/0
```

Router-Lisboa# show ip route

```
10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
O    10.10.1.0/24 [110/65] via 10.10.2.2, 00:02:36, Serial0/1/0
C    10.10.2.0/24 is directly connected, Serial0/1/0
L    10.10.2.1/32 is directly connected, Serial0/1/0
C    10.10.3.0/24 is directly connected, Serial0/1/1
L    10.10.3.1/32 is directly connected, Serial0/1/1
O    10.10.4.0/24 [110/65] via 10.10.3.2, 00:01:57, Serial0/1/1
```

Router-Porto# show ip route

```
10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
O IA  10.10.1.0/24 [110/129] via 10.10.3.1, 00:00:09, Serial0/1/0
O IA  10.10.2.0/24 [110/128] via 10.10.3.1, 00:00:09, Serial0/1/0
C    10.10.3.0/24 is directly connected, Serial0/1/0
L    10.10.3.2/32 is directly connected, Serial0/1/0
C    10.10.4.0/24 is directly connected, GigabitEthernet0/0/0
L    10.10.4.4/32 is directly connected, GigabitEthernet0/0/0
```

4 – Verifying the OSPF configuration

Router-Faro# show ip protocols

```
Routing Protocol is "ospf 100"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 10.10.2.2
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.1.0 0.0.0.255 area 0
    10.10.2.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.10.2.2             110          00:03:33
    10.10.3.1             110          00:03:03
  Distance: (default is 110)
```

Router-Lisboa# show ip protocols

```
Routing Protocol is "ospf 100"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 10.10.3.1
  Number of areas in this router is 2. 2 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.2.0 0.0.0.255 area 0
    10.10.3.0 0.0.0.255 area 1
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.10.2.2             110          00:04:29
    10.10.3.1             110          00:03:49
    10.10.4.4             110          00:03:49
  Distance: (default is 110)
```

Router-Porto# show ip protocols

```
Routing Protocol is "ospf 100"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 10.10.4.4
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.3.0 0.0.0.255 area 1
    10.10.4.0 0.0.0.255 area 1
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.10.3.1             110          00:04:55
    10.10.4.4             110          00:04:55
  Distance: (default is 110)
```

Other configuration commands to test:

Router-XXXXX# **show ip ospf**

apresenta um resumo da informação associada ao OSPF, nomeadamente os processos OSPF, a identificação do router, as áreas, estatísticas, etc.

Router-XXXXX# **show ip ospf database**

apresenta a base de dados do estado das ligações que formam a topologia de rede.

Router-XXXXX# **show ip ospf interface**

apresenta informação OSPF relativa às interfaces

Router-XXXXX# **show ip ospf neighbor**

apresenta a lista de vizinhos OSPF e o estado das adjacências

Pode verificar as mensagens de construção de adjacências em tempo real usando o comando de debug:

Router-XXXXX# **debug ip ospf adj**

Router-XXXXX# **no debug ip ospf adj**

5 – Verification of connectivity from machine A1

Connectivity of the A1 machine with its own network interface

C:\> **ping 10.10.1.1**

Pinging 10.10.1.1 with 32 bytes of data:

Reply from 10.10.1.1: bytes=32 time >10ms TTL=128

Reply from 10.10.1.1: bytes=32 time >10ms TTL=128

Reply from 10.10.1.1: bytes=32 time >10ms TTL=128

Reply from 10.10.1.1: bytes=32 time >10ms TTL=128

Connectivity between machine A1 and machine B1

C:\> **ping 10.10.4.1**

Pinging 10.10.4.1 with 32 bytes of data:

Reply from 10.10.4.1: bytes=32 time >10ms TTL=128

Reply from 10.10.4.1: bytes=32 time >10ms TTL=128

Reply from 10.10.4.1: bytes=32 time >10ms TTL=128

Reply from 10.10.4.1: bytes=32 time >10ms TTL=128

Connectivity between machine A1 and machine B2

C:\> **ping 10.10.4.2**

Pinging 10.10.4.2 with 32 bytes of data:

Reply from 10.10.4.2: bytes=32 time >10ms TTL=128

Reply from 10.10.4.2: bytes=32 time >10ms TTL=128
Reply from 10.10.4.2: bytes=32 time >10ms TTL=128
Reply from 10.10.4.2: bytes=32 time >10ms TTL=128

Connectivity between machine A1 and machine B3

C:\> **ping 10.10.4.3**
Pinging 10.10.4.2 with 32 bytes of data:
Reply from 10.10.4.3: bytes=32 time >10ms TTL=128
Reply from 10.10.4.3: bytes=32 time >10ms TTL=128
Reply from 10.10.4.3: bytes=32 time >10ms TTL=128
Reply from 10.10.4.3: bytes=32 time >10ms TTL=128

6 – Scenario Configuration with Stub Area

Consider the scenario from the previous configuration where you want to configure area 1 as a stub. To configure a router as belonging to an area of this type, use the command `area num_area stub`. Note that since the Faro router is internal to area 0 (or 0.0.0.0), it cannot be configured as a stub.

Finally, run the appropriate commands to confirm the new OSPF configuration and note the differences found.

Faro router configuration

```
Router-Faro> en  
Router-Faro# conf t  
Router-Faro (config)# router ospf 100  
Router-Faro (config-router)# network 10.10.1.0 0.0.0.255 area 0  
Router-Faro (config-router)# network 10.10.2.0 0.0.0.255 area 0  
Router-Faro (config-router)# ^Z
```

Porto router configuration

Em seguida, configura-se o router do Porto que segue uma estrutura semelhante.

```
Router-Porto > en  
Router-Porto # conf t  
Router (config)# router ospf 100  
Router-Porto (config-router)# network 10.10.3.0 0.0.0.255 area 1  
Router-Porto (config-router)# network 10.10.4.0 0.0.0.255 area 1  
Router-Porto (config-router)# area 1 stub  
Router-Porto (config-router)# ^Z
```

Lisboa router configuration

And finally, the Lisbon router, where the only difference from the others is that it has an interface with each of the areas.

```
Router-Lisboa > en  
Router-Lisboa # conf t  
Router-Lisboa (config)# router ospf 100  
Router-Lisboa (config-router)# network 10.10.2.0 0.0.0.255 area 0  
Router-Lisboa (config-router)# network 10.10.3.0 0.0.0.255 area 1  
Router-Lisboa (config-router)# area 1 stub  
Router-Lisboa (config-router)# ^Z
```

7 – Scenario Configuration with Totally Stub Area

As we saw in the theoretical sessions, summarized routes are not injected into totally stub areas.

- An OSPF stub area is an area that does not permit flooding of AS external link state advertisements.
- It only receives routing information for destinations within the same stub area and for destinations in other areas within the OSPF domain.
- To reach destinations external to the OSPF domain, default routes (imported by the area border router) are used

Therefore, it is up to the ABR router to block this type of route with the command `area number area stub no-summary`. In relation to the previous case, the ABR router is the Lisbon router, so it is only here that we need to enter the command to block summarized routes for area 1. As a result, area 1 is configured as totally stub.

```
Router-Lisboa# conf t  
Router-Lisboa (config)# router ospf 100  
Router-Lisboa (config-router)# network 10.10.2.0 0.0.0.255 area 0  
Router-Lisboa (config-router)# network 10.10.3.0 0.0.0.255 area 1  
Router-Lisboa (config-router)# area 1 stub no-summary
```

Finally, run the appropriate commands to confirm the new OSPF configuration and note any differences found, particularly regarding the routing table of the Porto router (within the stub area), where you should find the default route that the stub area defines for routing external routes to the OSPF system.

```

10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
O IA 10.10.1.0/24 [110/129] via 10.10.3.1, 00:37:05, Serial0/1/0
O IA 10.10.2.0/24 [110/128] via 10.10.3.1, 00:37:05, Serial0/1/0
C 10.10.3.0/24 is directly connected, Serial0/1/0
L 10.10.3.2/32 is directly connected, Serial0/1/0
C 10.10.4.0/24 is directly connected, GigabitEthernet0/0/0
L 10.10.4.4/32 is directly connected, GigabitEthernet0/0/0
O*IA 0.0.0.0/0 [110/65] via 10.10.3.1, 00:36:55, Serial0/1/0

```

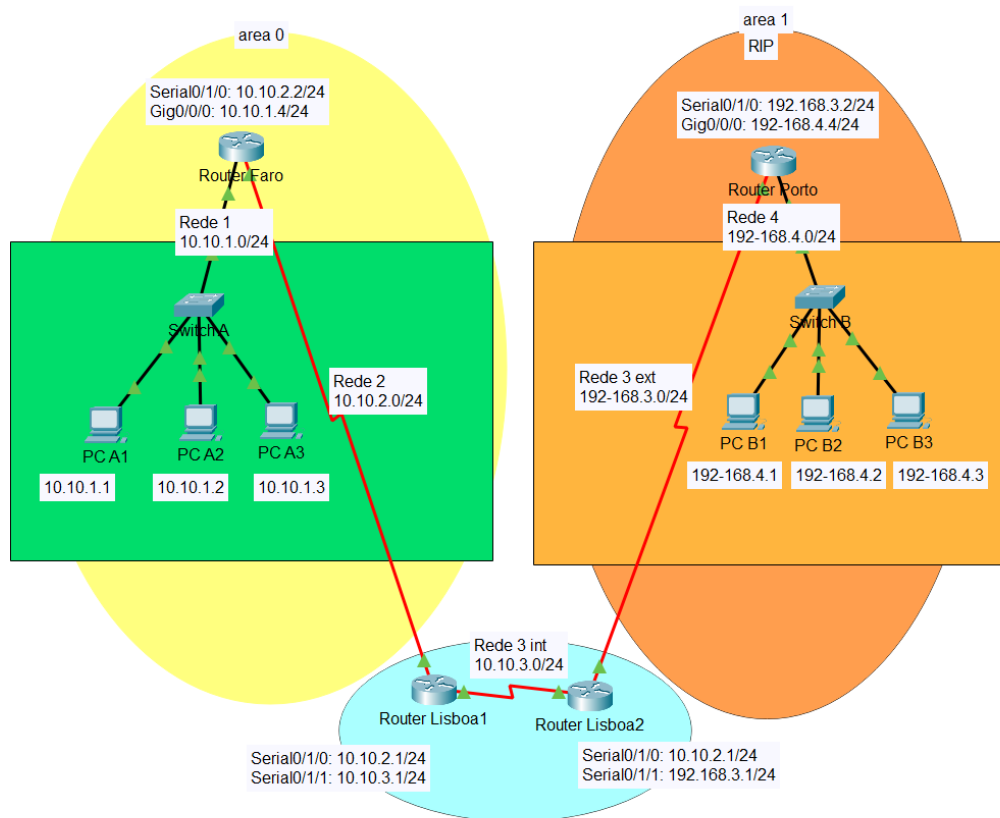
8 – Setting Up the Not-So-Stubby Area

To implement this scenario, we will use a variant of the initial configuration, where we now have two OSPF areas and one external area. The Porto router will be associated with an external network running RIP, area 0 will continue to be associated with the Faro router, and area 1 will now have two routers in Lisbon that are ABRs with the areas described above. This version of the topology can be found in the attached script with the suffix “_nssa.”

Not so stubby areas allow the injection of external routes while blocking LSA type 5.

Not So Stubby Area (NSSA): is a non proprietary extension of the existing stub area feature that allows the injection of external routes in a limited fashion into the stub area.

- Redistribution into an NSSA creates a special type of LSA known as a type 7 LSA, which can exist only in an NSSA.
- An NSSA ASBR generates this LSA, and an NSSA ABR translates it into a type 5 LSA, which gets propagated into the OSPF domain.
- The intention is that they do not pass LSA type 5 from area 0 to area 1, but that they can pass external RIP routes in area 1. The solution for this case is precisely to configure area 1 as not-so-stubby. This new topology can be obtained at the following link, and you should check the configurations made previously.



Finally, run the appropriate commands to confirm the new OSPF configuration and note any differences found. (show ip ospf database, show ip protocols)

Faro router configuration

The Faro router belongs to area 0 and has no connection to the NSSA area, so its configuration has no points associated with NSSA areas.

```
Router-Faro > en
Router-Faro# conf t
Router-Faro (config)# router ospf 100
Router-Faro (config-router)# network 10.10.0.0 0.0.255.255 area 0
Router-Faro (config-router)# ^Z
```

Porto router configuration

Next, configure the Porto router, which, in this topology, is an internal router in an autonomous RIP system, so it has no NSSA-related configuration.

```
Router-Porto > en
Router-Porto # conf t
Router (config)# router rip
```

```
Router-Porto (config-router)# network 192.168.3.0
```

```
Router-Porto (config-router)# network 192.168.4.0
```

```
Router- Porto (config-router)# ^Z
```

Lisboa router configuration

Router 1 in Lisbon is a border router with an NSSA area (area 1), so it is necessary to indicate that this router belongs to an NSSA area using the command `area area_number nssa`. Next, router 2 in Lisbon also belongs to the NSSA area, so it must be configured with a command identical to the previous one. In addition, as it participates in two routing protocols, OSPF and RIP, it is necessary to redistribute routes between the two protocols. To do this, it is necessary to indicate that you want to redistribute RIP (`redistribute rip`) and OSPF (`redistribute ospf 100`). The `default-metric` value command is used to indicate the metric to be associated with the routes that will be redistributed accordingly.

```
Router1-Lisboa (config)# router ospf 100
```

```
Router1-Lisboa (config-router)# network 10.10.2.0 0.0.0.255 area 0
```

```
Router1-Lisboa (config-router)# network 10.10.3.0 0.0.0.255 area 1
```

```
Router1- Lisboa (config-router)# area 1 nssa
```

```
Router1- Lisboa (config-router)# ^Z
```

```
Router2-Lisboa (config)# router ospf 100
```

```
Router2-Lisboa (config-router)# redistribute rip metric 110 subnets
```

```
Router2-Lisboa (config-router)# network 10.10.3.0 0.0.0.255 area 1
```

```
Router2-Lisboa (config-router)# area 1 nssa
```

```
Router2-Lisboa (config)# router rip
```

```
Router2-Lisboa (config-router)# redistribute ospf 100 metric 10
```

```
Router2-Lisboa (config-router)# network 10.10.3.0
```

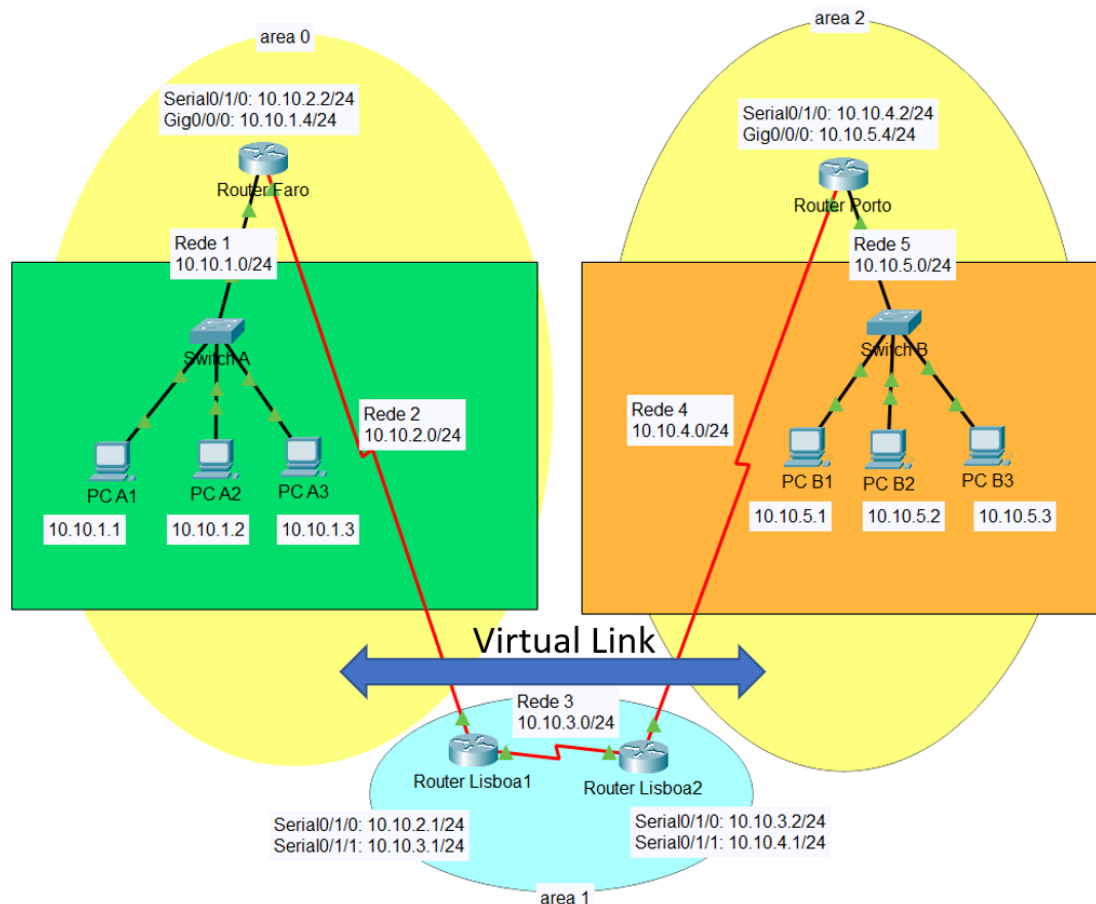
```
Router2-Lisboa (config-router)# network 10.10.4.0
```

Supplementary questions:

- 1) Check the LSA types circulating in area 0 using the `ip ospf database` command on the Faro router.
- 2) Check that LSA type 5 does not circulate in the NSSA area (area 1).
- 3) Use the `show ip protocols` command on each of the routers and comment on the differences.
- 4) Check the connectivity between the machines.

9 – Virtual Links in OSPF

To carry out this phase of the work, a variation of the base topology will be considered, where there will be three OSPF areas: area 0 (Faro router), area 1 (router 1 and router 2 in Lisbon), and area 2 (Porto router). Addressing adjustments will also be considered, as illustrated in the following figure. This topology can be obtained via the following link, and you should check the configurations previously made.



As mentioned in the theory section, all OSPF areas must be connected to area 0. Whenever it is not possible to establish a physical connection between an area and the backbone area, it is necessary to create a virtual connection that maintains the validity of the rule.

The virtual link is established between the ABR routers that connect area 0 and the transition area, and between another area and the transition area. To create the virtual link, use the command `area num_area virtual-link id_router` applied to the ABR routers, where `num_area` identifies the intermediate area and `id_router` is the highest loopback IP address of the router or, if there is no loopback interface, it will be the highest configured IP address.

Router configuration

The routers in Lisbon will have to be configured to create a virtual connection. Let's just consider the configuration between these, since the others are internal routers within their respective OSPF areas and do not intervene in the process.

Faro router configuration

The Faro router is internal to area 0 and does not intervene in the virtual connection.

```
Router-Faro > en
Router-Faro # conf t
Router-Faro (config)# router ospf 100
Router-Faro (config-router)# network 10.10.0.0 0.0.255.255 area 0
Router-Faro (config-router)# ^Z
```

Porto router configuration

The Porto router, which is an internal router in area 2, also does not intervene in the virtual connection.

```
Router-Porto > en
Router-Porto # conf t
Router (config)# router ospf 100
Router-Porto (config-router)# network 10.10.4.0 0.0.0.255 area 2
Router- Porto (config-router)# network 10.10.5.0 0.0.0.255 area 2
Router- Porto (config-router)# ^Z
```

Lisboa router configuration

The routers in Lisbon will have to be configured to create a virtual connection. It should be noted that the virtual connection considers the addresses of interfaces 10.10.3.1 and 10.10.4.1, which are the largest addresses configured on the routers in Lisbon, taking into account that there are no loopback addresses.

```
Router1-Lisboa (config)# router ospf 100
Router1-Lisboa (config-router)# network 10.10.2.0 0.0.0.255 area 0
Router1-Lisboa (config-router)# network 10.10.3.0 0.0.0.255 area 1
Router1- Lisboa (config-router)# area 1 virtual-link 10.10.4.1
Router1- Lisboa (config-router)# ^Z
```

```
Router2-Lisboa (config)# router ospf 100
Router2-Lisboa (config-router)# network 10.10.3.0 0.0.0.255 area 1
Router2-Lisboa (config-router)# network 10.10.4.0 0.0.0.255 area 2
Router2- Lisboa (config-router)# area 1 virtual-link 10.10.3.1
```

Supplementary questions

- 1) Check neighbor relationships using the `show ip ospf neighbor` command
- 2) Confirm the virtual link configuration using the `show ip ospf virtual-links` command
- 3) Check connectivity between machines.